

SARWAN KUMAR

GENAI & BACKEND SYSTEMS ENGINEER

6283202698

@sarwankrchaubey

sarwanchaubey@zohomail.in

Chandigarh

SUMMARY

Experienced Backend & GenAI Systems Engineer with 5 years of experience building scalable microservices, cloud-native systems, and AI-powered applications. Strong expertise in Node.js (TypeScript), distributed backend architecture, and AWS serverless infrastructure, with a proven track record of designing systems capable of handling millions of requests and large-scale data processing workloads.

Hands-on experience integrating **LLM-powered applications**, vector search pipelines, and AI services such as **AWS Bedrock**, along with implementing **parameter-efficient model fine-tuning techniques (LoRA/PEFT)** using **HuggingFace Transformers**. Skilled in designing **RAG architectures**, high-performance APIs, and real-time data processing pipelines using technologies like MongoDB, MySQL, and scalable cloud services.

Strong background in system performance optimization, microservices architecture, and large-scale data querying, including advanced MongoDB aggregation pipelines and database tuning for high-throughput, low-latency systems. Experienced in building production-grade cloud systems following 12-Factor App principles and leading engineering teams to deliver reliable, scalable backend platforms.

TECHNICAL SKILLS

Core Languages	Backend & DB	AI / GenAI	Cloud & Infrastructure
- Node.js - Javascript - TypeScript - Python	- NestJs - REST API - MongoDB - PostgreSQL - Microservices - GraphQL	- LLM Applications - LangChain - LangGraph - LoRA - Vector DB - AWS Bedrock	- AWS Lambda - API Gateway - Serverless - S3 , EC2 - Opensearch - Docker

WORK EXPERIENCE

SOFTWARE ENGINEER(FULL STACK) (PROJECT TEAM LEAD)

Orion Solution

(Aug 2024 - Present)

- Designed and implemented CI/CD pipelines using GitHub Actions, automating build, test, and deployment workflows for Node.js-based microservices. This improved deployment speed by 30% and supported scalable delivery through robust branching strategies.
- Developed robust backend systems using Nodejs, MySQL , **GraphQL** and AWS, implementing asynchronous programming and multithreading techniques to optimize processing performance and handle concurrent tasks effectively, improving API response times by **25%**.
- Applied asynchronous techniques to batch process over 3 million data records, achieving a 40% reduction in processing time and significantly improving real-time data handling.
- Implemented thread synchronization and resource management to prevent race conditions, increasing application stability under heavy workloads and **improving throughput by 15%**.
- Adopted the 12-Factor App methodology for building modern, maintainable, and scalable web applications.
- Led a cross-functional team of 6 developers** (frontend and backend), overseeing system **architecture design, tech stack selection**, task planning, and **sprint execution**. Facilitated seamless collaboration, code reviews, and mentorship, ensuring on-time delivery of scalable and high-quality software solutions.
- Designed and implemented **LLM-powered backend** capabilities, including **Retrieval-Augmented Generation (RAG) pipelines** using **vector databases**, enabling semantic search and AI-assisted data retrieval for large datasets.
- Built **LLM fine-tuning pipeline** using **HuggingFace Transformers and LoRA (PEFT)**, preparing instruction datasets and training domain-specific models for backend AI services.

SENIOR BACKEND DEVELOPER

Rebuild Technologies

(Jan 2024 - Aug 2024)

- Broke down the monolithic application into independent microservices, significantly increasing the scalability of the system and enabling it to handle up to **5 million requests**.
- Enabled more frequent updates and faster feature releases, accommodating a **50% increase in deployment** cadence.
- Established CI/CD pipelines using GitHub Actions to automate the build, testing, and deployment processes for each microservice. This automation streamlined workflows, reducing manual intervention and improving **productivity by 30%**.

- Enhanced **database by 25% due to the implementation of connection pooling**, reducing average response time to 150ms per query.
- **Led a team of 5 developers** across **multiple microservice projects**, handling **code reviews**, **cloud deployments (AWS)**, and inter-service communication while ensuring code quality and timely delivery.

SOFTWARE DEVELOPER

(2022 - 2024)

Zoptal Solution Pvt Limited

- Designed and implemented scalable RESTful APIs using AWS Lambda and API Gateway, leveraging AWS serverless architecture to build cost-efficient and highly available backend services. Integrated backend APIs with React.js applications and implemented efficient state management using Redux to support reliable data flow across the system.
- Optimizing application performance, scalability, and reliability by using MongoDB aggregation Pipeline.

SOFTWARE DEVELOPER

(2021 - 2022)

HANSRAJ CONSULTANCY

- Built responsive and visually appealing front-end interfaces using React.js through personal and academic projects.
- Optimizing application performance, scalability, and reliability by using MongoDB aggregation Pipeline.
- Developed RESTful APIs to connect front-end applications with backend services.
- Implemented AWS Lambda and API Gateway to understand serverless application development.
- Build Redux store for managing state in React applications.
- Explored AWS serverless architecture as part of self-learning to understand scalable system design.
- Worked with MongoDB Aggregation Pipeline to perform efficient data processing and querying in project work.

PROJECTS

RECRUITMENT PORTAL - KINGSLEYGATE

- The project is a HR portal where the companies can mark, evaluate and contact prospective talent
 - **TECH STACK**
 - Node.js (NestJS), AWS Lambda, API Gateway, Vector Database, AWS Bedrock, LangChain
 - **My Role**
 - Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline to enable AI-assisted candidate search and knowledge retrieval across large HR datasets.
 - Built vector-based semantic search pipelines, embedding candidate profiles and HR documents into a vector database to support contextual AI queries.
 - Integrated LLM inference using **Amazon Bedrock models**, enabling natural language candidate discovery and automated HR insights.
 - Developed a **LoRA-based** model fine-tuning pipeline using **HuggingFace Transformers**, preparing instruction datasets and training domain-specific models to improve response quality. (POC)

REAL-ESTATE LISTING WEBSITE - MUNZUL DEV

- The project has a customer-side application and an admin panel.
 - **TECH STACK**
 - Backend : Express , NodeJs
 - Customer Application : NextJs
 - Admin Panel : NextJs
 - **My Role**
 - **Backend Development:** Responsible for developing the server-side business logic using Springboot, ensuring efficient handling of requests, and maintaining a scalable and secure architecture.
 - **Front-End Development:** Developed the front-end for both the customer application and the admin panel using Next.js. This included designing and implementing user interfaces, managing state, and integrating APIs.
 - **Database Management:** Utilized both MySQL and MongoDB depending on the use case:
 - MongoDB: Used for storing non-relational, random data that required flexibility and scalability.
 - **Panel Development:** Although the main focus was on the customer application and admin panel, also contributed to the development of the merchant panel and business panel, ensuring seamless interaction between different components of the system.

EDUCATION AND CERTIFICATION

BACHELOR OF TECHNOLOGY (ELECTRICAL & ELECTRONICS ENGINEERING)

(2015 - 2019)

Lovely Professional University